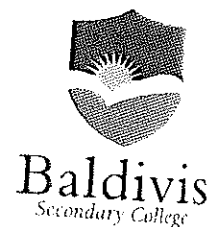


YEAR 8 Algebra
Semester 1 2017
Revision **MOCK** Test 2



Name: Solutions

Total Marks: 55 marks
Total Time: 55 minutes

**Full working out must be shown to get full marks.
Attempt all questions**

Resources allowed:
1 A4 page, (1 side) of hand written notes
Calculator

1. For the expression $4xy + 3x - 6y^2 + 2$, state:
(4)

- a) the **coefficient** of x 3
- b) the **constant** term 2
- c) the number of **terms** 4
- d) the coefficient of xy 4

2. State whether each of the following is **true** or **false**:
(5)

- a) The constant term in the expression $7x + 5$ is 7. F (= 5)
- b) $15xy$ and $6yx$ are like terms. T
- c) The coefficient of d in the expression $6d^2 + 4d + 8abd + 3$ is 4. T
- d) The highest common factor **HCF** of $12bac$ and $16c$ is $2c$. F (= $4c$)
- e) The coefficient of x and y in the expression $2x + 2y$ is 2. T

3. Circle the **like terms** in each group
(5)

a) 2m, 3n, m

b) 9jf, 3f, -2f

c) $10d, 3d, 9e$

e) $-6w, 4q, -2w$

d) $14s^2, 14s, 2s^2$

4. Simplify the following. .
(4)

a) $3a + 3a = 6a$

c) $24d - 15d = 9d$

b) $19i - 4i = 15i$

d) $2g - g + 4g = 5g$

5. Simplify the following. .
(4)

a) $14c + 6p + 3c = 17c + 6p$

c) $-4b + 6h + 3b + 2h = 8h - b$

b) $8f + 6f - 9j = 14f - 9j$

d) $3q - 7q + 4j + 5j = 9j - 4q$

6. Simplify:

a) $10a \times 4b = 40ab$

c) $8m \times -n \times 4 = -32mn$

b) $2pq \times 6 \times r = 12pqr$

d) $4a \times a^2 = 4a^3$

7. Simplify the following.
(4)

a) $4a \times 2bc \times 7d = 56abcd$

b) $4xy \times 4y = 16xy^2$ or $4xxy^2$

$$c) \frac{5 \frac{25ef}{5e}}{1} = \frac{5f}{1} = 5f$$

$$d) \frac{20x^2y}{36xz} = \frac{\cancel{4} \times 5 \times \cancel{x} \times \cancel{x} \times y}{\cancel{4} \times 9 \times \cancel{z} \times 2} = \frac{5xy}{9z}$$

8. Expand:

(3)

$$a) 5(2 + x) \quad 5 \times 2 + (5 \times x)$$

$$b) x(10y + 1) \quad (x \times 10y) + (x \times 1)$$

$$c) 3x(2y + 5)$$

$$(3x \times 2y) + (3x \times 5)$$

9. Simplify by expanding and collecting like terms:

(4)

$$a) \quad 7(a + 2) + 3a$$

$$3(2a - ab)$$

$$7a + 14 + 3a$$

$$= \underline{\underline{10a + 14}}$$

$$b) a(3 + 2b) + 3(2a - ab)$$

$$3a + 2ab + 6a - 3ab$$

$$\underline{\underline{9a - ab}}$$

10. Evaluate the following if "a" = 5 and "b" = 3

(4)

$$a) a + b$$

$$5 + 3 = \underline{\underline{8}}$$

$$b) 10a + 7b$$

$$10(5) + 7(3)$$

$$50 + 21 = \underline{\underline{71}}$$

$$c) -5a^2$$

$$-5(5)^2$$

$$= \underline{\underline{-125}}$$

$$d) a^2 - b^2$$

$$5^2 - 3^2$$

$$= 25 - 9$$

$$= \underline{\underline{16}}$$

11. Evaluate the following expressions when $d = -2$ and $e = 5$

(4)

a) $3de - 6$

$$\begin{aligned} 3(-2)(5) - 6 \\ 3(-10) - 6 \\ -30 - 6 = \underline{\underline{-36}} \end{aligned}$$

c) $e^2 - 3d^2$

$$\begin{aligned} 5^2 - 3(-2)^2 \\ 5^2 - 3(4) \quad 25 - 12 \\ 5^2 - 12 = \underline{\underline{13}} \end{aligned}$$

12. Factorise fully.

(4)

a) $2x + 6$

$$2(x + 3)$$

c) $24 - 20g$

$$4(6 - 5g)$$

b) $5de + 100$

$$\begin{aligned} 5(-2)(5) + 100 \\ 5(-10) + 100 \\ -50 + 100 = \underline{\underline{50}} \end{aligned}$$

d) $2e^3 - 3d^2$

$$\begin{aligned} 2(5)^3 - 3(-2)^2 \\ 2(125) - 3(4) \\ 250 - 12 \\ = \underline{\underline{238}} \end{aligned}$$

b) $7a^2 + 21ab$

$$7a(a + 3b)$$

d) $12x + 4xy$

$$4x(3 + y)$$

13. Two bus companies have different pricing structures.

(6)

Company A: \$100 call-out fee, plus \$60 per hour

Company B: \$70 call-out fee, plus \$90 per hour

a) Write an expression for the total cost \$A of travelling n hours with company A.

$$TC(A) = \$60n + \$100$$

b) Write an expression for the total cost \$B of travelling for n hours with company

$$TC(B) = \$90n + \$70$$

To check,
re expand....

- c) Hence state the cost of travelling for 3 hours with each company.

$$A = \$60 \times 3 + \$100 = \$280$$

$$B = \$90 \times 3 + \$70 = \$340$$

- d) For how long would you need to hire a bus to make company A the cheaper option?

At 2 hours. (see Table of Values Below)

At \$70 flagfall B is cheaper, then same.

- e) In the end, a school principal cannot decide which bus company to choose and hires 5 buses from company A and 4 buses from company B. Give an expanded expression to find the total cost for the school to hire the five buses for n hours.

$$TC = 5(\$60n + \$100) + 4(\$90n + \$70)$$

- f) If the trip lasts for 6 hours, how much does it cost to hire the nine buses for this period of time?

$$\begin{aligned} TC &= 5(\$60 \times 6 + \$100) + 4(\$90 \times 6 + \$70) \\ &= 5(460) + 4(610) \\ &= 2440 + 2300 \\ &= \underline{\underline{4740}} \end{aligned}$$

T.O.V.

A	x	c	1	2	3
	y	100	160	220	280
B	x	c	1	2	3
	y	70	160	250	

